LINEAR ALGEBRA · LESSON 7

Solving Ax = 0

Which trades change **nothing**? Simplify $A \to U \to R$, then read every answer off R.

Rank, pivot and free variables, special solutions.

A → U → R | n – r free | MIT 18.06, Lecture 7

THE QUESTION

Four funds, three rules

Nifty, Nifty ETF (its twin), bonds, balanced (half Nifty, half bonds). Rows: budget, return, exposure.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix}$$

Two do-nothing trades jump out: swap Nifty for its twin, and build balanced from Nifty + bonds. **Are there more?**

STEP 1 · A → U

Clear below the pivots

$$U = \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & 0 & \boxed{-5} & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Row 2 − 12 × row 1 asks: how much return **beyond** 12%? Row 3 ends up all zeros: exposure told us nothing new.

Safe: a trade that answers 0 to the old rules answers 0 to the new ones.

STEP 2 · U → R

Pivots to 1, clear above

Row 2 ÷ (−5), then row 1 − row 2:

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 \\ 0 & 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Row 1: $x_1 + x_2 + 0.5x_4 = 0$. Row 2: $x_3 + 0.5x_4 = 0$. Each rule now involves **one** pivot amount only.

THE RANK

r = number of pivots = 2

- **2 of 3 rules** carry information
- **2 of 4 funds** bring something new (Nifty, bonds)
- **4 − 2 = 2** amounts are left free

PIVOT AND FREE

You pick, R decides

Free (no pivot): ETF, balanced. **Pivot:** Nifty, bonds.

Pick ETF = 3, balanced = 2. Read R:

$$\text{bonds} = -0.5(2) = -1, \quad \text{Nifty} = -3 - 0.5(2) = -4$$

> COMMON SENSE AGREES
>
> Undo the ETF by selling 3 Nifty; undo balanced by selling 1 Nifty + 1 bonds.

TWO CASES

When is there a non-zero solution?

Menu	r	free	Non-zero?
Nifty, bonds	2	0	no: only 0
Nifty, bonds, midcap	3	0	no: only 0
Nifty, bonds, balanced	2	1	yes, ∞
Nifty, ETF, bonds, balanced	2	2	yes, ∞

A free column ⇒ yes. More funds than rules ⇒ always yes.

Read them off R

One free amount = 1, the other 0. Pivot amounts = **minus** that column of R:

- ETF = 1: column $(1, 0) \to s_1 = (-1, 1, 0, 0)$: guess 1
- Balanced = 1: column $(0.5, 0.5) \to s_2 = (-0.5, 0, -0.5, 1)$: guess 2

Every do-nothing trade is $c\,s_1 + d\,s_2$, and nothing else exists.

IN MARKETS

Four products, three scenarios

Futures, 20,000 call, 20,000 put, 19,900 call. R has one free column (the put) and one special solution:

$$(1, -1, 1, 0)$$

Long future, short call, long put pays 0 everywhere: **put–call parity**.

RECAP

Key takeaways

1. $A \to U \to R$ keeps the do-nothing trades.
2. Rank r = number of pivots.
3. Pivot amounts are decided by R; free ones you choose.
4. Free column ⇒ non-zero solutions; none ⇒ only 0.
5. Special solution: one free = 1, pivots = −column of R.
6. All solutions: mixes of the special solutions.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A lab that takes any matrix up to 3 × 5 from A to U to R and finds every special solution, plus worked hedging and options examples:

learn.krisarj.com

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